

Conditionals

Victor Eijkhout, Susan Lindsey

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1. If-then-else

A conditional is a test: 'if something is true, then do this, otherwise maybe do something else'. The C++ syntax is

```
1 if ( something ) {  
2   // do something;  
3 } else {  
4   // do otherwise;  
5 }
```

- The 'else' part is optional
- You can leave out braces in case of single statement.

2. Complicated conditionals

Chain:

```
1 if ( /* some test */ ) {  
2     ...  
3 } else if ( /* other test */ ) {  
4     ...  
5 }
```

Nest:

```
1 if ( /* some test */ ) {  
2     if ( /* other test */ ) {  
3         ...  
4     } else {  
5         ...  
6     }  
7 }
```

3. Comparison and logical operators

Here are the most common logic operators and comparison operators:

Operator	meaning	example
==	equals	<code>x==y-1</code>
!=	not equals	<code>x*x!=5</code>
>	greater	<code>y>x-1</code>
>=	greater or equal	<code>sqrt(y)>=7</code>
<,<=	less, less equal	
&&,	and, or	<code>x<1 && x>0</code>
and,or	and, or	<code>x<1 and x>0</code>
!	not	<code>!(x>1 && x<2)</code>
not		<code>not (x>1 and x<2)</code>

Precedence rules of operators are common sense. When in doubt, use parentheses.

Exercise 1

The following code claims to detect if an integer has more than 2 digits.

Code:

```
1 // basic/if.cpp
2 int i;
3 cin >> i;
4 if ( i>100 )
5     println
6     ("Number {} had more than 2
7     digits",
8     i);
```

Output:

```
1 ... with 50 as input
  ↪....
2 ... with 150 as
  ↪input ....
3 Number 150 had more
  ↪than 2 digits
```

Fix the small error in this code. Also add an 'else' part that prints if a number is negative.

4. About cout and print

```
1 #include <iostream>
2 using std::cout
3
4 std::string world="world";
5 cout << "Hello " << world << '\n';
```

```
1 #include <print>
2 using std::println;
3
4 println( "Hello {}",world );
```

- `cout` is somewhat clumsy
- `print` and `println` are more modern, more concise, more flexible.
- Big caveat: needs C++23:

```
icpx -std=c++23 yourfile.cpp
```

Quiz 1

True or false?

1. The tests `if (i>0)` and `if (0<i)` are equivalent.
2. The test

```
1 if (i<0 && i>1)
2   cout << "foo"
```

prints foo if $i < 0$ and also if $i > 1$.

3. The test

```
1 if (0<i<1)
2   cout << "foo"
```

prints foo if i is between zero and one.

Quiz 2

Do you think this is good code?

```
1 bool x;  
2 // ... code that sets x ...  
3 if ( x == true )  
4     // do something
```


Exercise 2

Read in a positive integer. If it's a multiple of three print 'Fizz!'; if it's a multiple of five print 'Buzz!'. If it is a multiple of both three and five print 'Fizzbuzz!'. Otherwise print nothing.

Note:

- Capitalization.
- Exclamation mark.
- Your program should display at most one line of output.

Prime Project Exercise 3

Read two numbers and print a message stating whether the second is as divisor of the first:

Code:

```
1 // primes/divisiontest.cpp
2 int number,divisor;
3 bool is_a_divisor;
4 /* ... */
5 if (
6     /* ... */
7     ) {
8     println
9     ("Yes, {} is a divisor of {}",
10     divisor,number);
11 } else {
12     println
13     ("No, {} is not a divisor of {}"
14     ,
15     divisor,number);
16 }
```

Output:

```
1 ( echo 6 ; echo 2 )
   ↪| divisiontest
2 Enter a number:
3 Enter a trial
   ↪divisor:
4 Yes, 2 is a divisor
   ↪of 6
5
6 ( echo 9 ; echo 2 )
   ↪| divisiontest
7 Enter a number:
8 Enter a trial
   ↪divisor:
9 No, 2 is not a
   ↪divisor of 9
```

Scope of variables

5. Local variables in conditionals

The curly brackets in a conditional allow you to define local variables:

```
1 if ( something ) {  
2     int i;  
3     .... do something with i  
4 }  
5 // the variable `i' has gone away.
```

Good practice: only define variable where needed.

Braces induce a scope.

6. Conditional with initializer

Variable created and set local to the conditional:

Code:

```
1 // basic/ifinit.cpp
2 if ( char c = getchar(); c!='a' )
3     cout << "Not an a, but: "
4         << c << '\n';
5 else
6     cout << "That was an a!"
7         << '\n';
```

Output:

```
1 for c in d b a z ;
    ↪do \
2     echo $c |
    ↪./ifinit ; \
3 done
4 Not an a, but: d
5 Not an a, but: b
6 That was an a!
7 Not an a, but: z
```

(No worries if you don't understand the Unix scripting)

Switch statement

7. Switch statement example

Cases are evaluated consecutively until you 'break' out of the switch statement:

Code:

```
1 // basic/switch.cpp
2 switch (n) {
3 case 1 :
4 case 2 :
5     cout << "very small" << '\n';
6     break;
7 case 3 :
8     cout << "trinity" << '\n';
9     break;
10 default :
11     cout << "large" << '\n';
12 }
```

Output:

```
1 for v in 1 2 3 4 5 ; do \
2     echo $v |
3     ↪ ./switch ; \
4     done
4 very small
5 very small
6 trinity
7 large
8 large
```

(No worries if you don't understand the Unix scripting)